Michael Cusack-Nelkin

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Education

Columbia University | M.A. in Mathematics | New York, NY | Aug 2025 - Dec 2026

- Master's in the Mathematics of Finance
- GPA: 3.6 / 4.00
- GRE Score: 169V - 166Q - 4.5W

The College of William & Mary | B.S. in Physics | Williamsburg, VA | Aug 2018 - May 2022

- GPA: 3.7 / 4.00
- Minors in Data Science and Mathematics
- Honors: Dean's List, *magna cum laude*

Work Experience

Deloitte Consulting | Strategy and Analytics Analyst - State of Texas | Austin, TX | Oct 2022 - Jun 2025

- Led four developers and testers in delivering a fraud recovery system for the Texas Office of Inspector General, identifying and recovering $10 million in previously unrecoverable fraudulent Medicaid claims.
- Presented technical progress, design decisions, and testing results to a mixed audience of technical and non-technical clients multiple times per week, maintaining stakeholder support throughout development, implementation, and claim recovery.

DLA Piper | Financial Analyst Intern | New York, NY | Jun 2020 - Sep 2020

- Analyzed practice-level profitability and billing data. Prepared weekly reports for finance leadership, allowing them to better understand the profitability and performance of individual teams and clients.

Gilder, Gagnon, Howe & Co. | Research Analyst Intern | New York, NY | Mar 2018 - Jun 2018

- Analyzed public online gaming companies' business models, competitive position, and financials using 10-Ks, 10-Qs, and earnings calls.
- Participated in investment committee meetings and portfolio company interviews in agtech, biotech, and live entertainment.

Research and Competitions

Millennium x Columbia University Trading Competition | New York, NY

- Led a team of four to win a Millennium Management ETF arbitrage competition for Columbia students. Used a rolling z-score to estimate a hedging relationship between the ETF and its underlying components, and traded a synthetic basket of the underlyings and ETF.

High Sharpe, Low Capacity: Optimally Trading the Nasdaq Closing Auction Imbalance | Summer 2026

- Built a statistical arbitrage system combining a return forecast, 10-factor covariance model, spread costs, and price impact estimates into a constrained portfolio optimization problem in Python using CVXPY.
- Backtested the strategy over 110 days and 100 stocks. Estimated an 8.53 Sharpe ratio at a maximum capacity of $91.3 million GMV.
- Identified fragility in capacity and latency requirements that favor use within a larger portfolio's execution process.

Adaptive Execution Strategies Lead to Robust Improvements Over Naive Benchmark | Spring 2026

- Co-developed an execution simulator using L2 Nasdaq order book data, supporting market, resting limit, and IOC orders under configurable latency and stochastic replenishment assumptions.
- Designed an execution strategy that used book imbalance to adjust fill aggression and reduced simulated arrival costs by 55%-89% versus a naive benchmark across three studied assets.

Technical Skills

- Finance: financial statement analysis, strategic analysis, SEC filings, research writing, market structure, competitive analysis.
- Programming and Methods: Python, NumPy, Pandas, Polars, scikit-learn, statsmodels, XGBoost, CVXPY, Git, portfolio optimization, factor risk modeling, market microstructure analysis, data cleaning and normalization.